

Colliding Wind Binary with a Supernova Event in AMUSE: The Morphology of the Apep System

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1 Introduction and motivation

In the Norma constellation one can find the Apep system (J. R. Callingham, Crowther, et al. 2020): a colliding wind Wolf-Rayet binary showing a pinwheel-like stellar wind structure. In Figure 1, the Very Large Telescope (VLT) observation is shown:

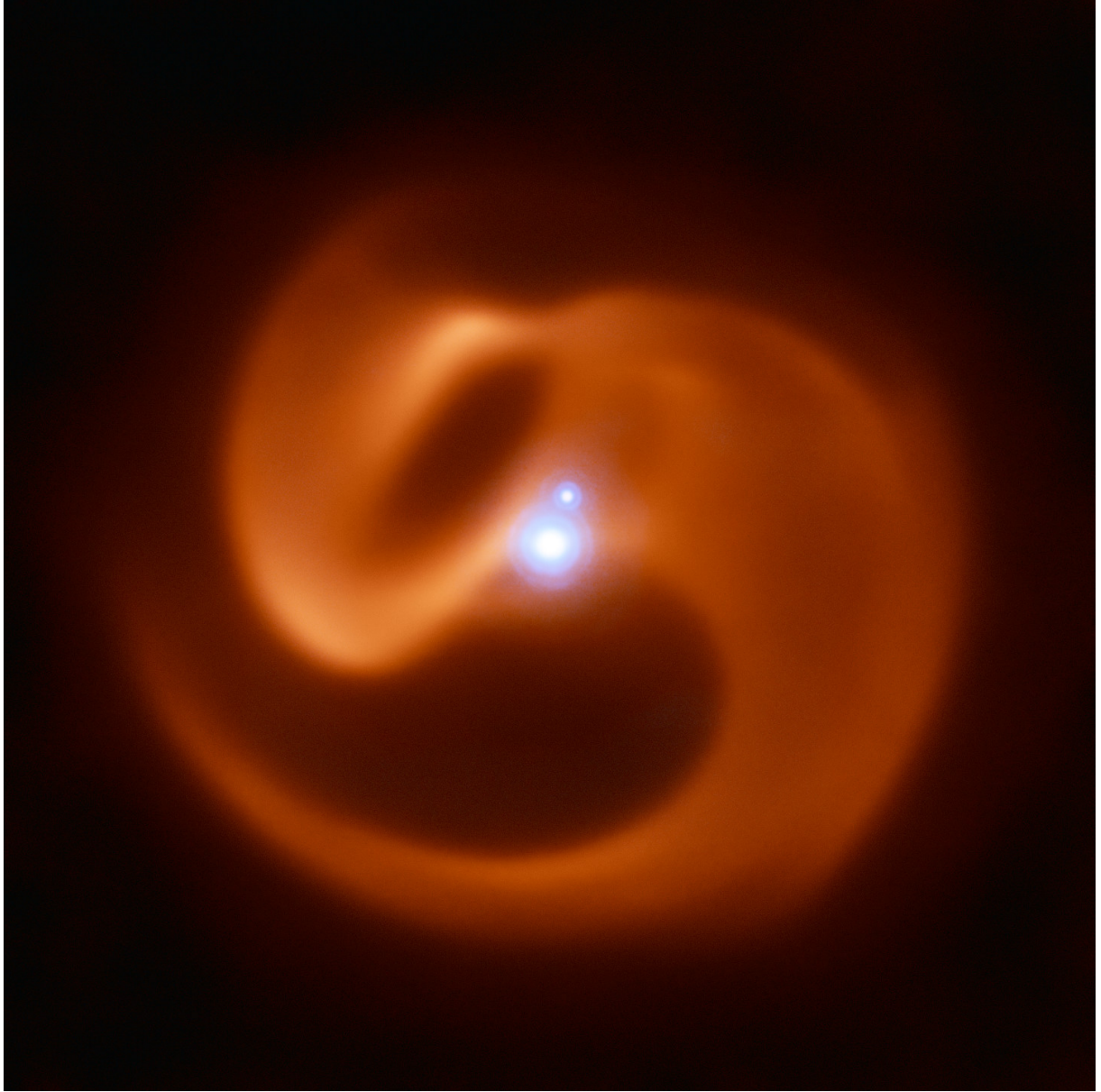


Figure 1: The Apep system as observed by the VISIR instrument on the Very Large Telescope (VLT). Credit ESO/Callingham et al.

The Apep system has a very peculiar morphology. Wolf Rayet binaries are rare, especially those exhibiting pinwheel or spiral structures in their colliding winds (J. R. Callingham, Crowther, et al. 2020; Han et al. 2020). How this shape has exactly come to be is still an unanswered question. Some suggest a near-critical rotation around its own axis of one of the Wolf-Rayet stars (Han et al. 2020), which would also make it the first type of long-duration gamma-ray burst (LGRB) progenitor observed (J. R. Callingham, Tuthill, et al. 2018). Wolf-Rayet (WR)

stars are in the last phase of evolution before going supernova, and this system in particular may also produce LGRBs. It is hard to prove this near-critical rotation of a WR star due to observational constraints. Theoretically, the rotation of one of the WR stars could be caused by binary interaction. This idea can be rejected based on the proximity of the binary to each other, causing them to be more or less tidally locked. However, in the wind and emission models produced by (J. R. Callingham, Tuthill, et al. 2018; Han et al. 2020; Palacio et al. 2022) they see that the near-critical rotation of one of the WR stars and the other star rotating very slowly is one of the main drivers – together with their ability to launch extreme anisotropic winds – behind obtaining the spiral shape that is observed in Figure 1. All in all, Apep is a unique and mysterious system that can teach us a lot about the evolution of stars in the Milky Way and the effects of stellar winds on their surrounding environment (and the interstellar medium).

In Figure 1 it might not be immediately obvious there is a colliding wind WR binary. The central light source is this WR binary, whilst 1700 au to the north is a supergiant. In an interview we had with Joe Callingham, Callingham told us that this supergiant likely formed there and orbits the 'Central Engine', i.e. the WR binary. The WR binary consists of a carbon-burning WR star and a nitrogen-burning WR star, which are separated by only 67 AU, according to the model from (Han et al. 2020). There is no evidence as to which of these WR stars would have the near-critical rotation needed to produce the spiral structure. Since the WR stars are in their final stage of evolution, they will go supernova in the foreseeable future. Callingham noted that the nitrogen-burning star will likely go supernova first. This will be very interesting to see, as it will likely have a large effect on the morphology of the colliding winds.

Many questions remain about Apep after its discovery and several studies on the astrophysics behind the colliding winds. Hence, it is worthwhile to continue studying this system with various approaches. The *James Webb Space Telescope* (JWST) and the *Atacama Large Millimeter/submillimeter Array* (ALMA) will improve the understanding of Apep significantly (J. Callingham 2019) with higher sensitivity and much better infrared capabilities than VLT. In (J. Callingham 2019) Callingham also mentions that detailed theoretical modeling will be key to understanding Apep, and this is where we come in. The Astrophysical Multipurpose Software Environment (AMUSE) is a useful tool for simulating astrophysical systems using well-known physics and clever computing techniques. In this report, we discuss our simulation of the Apep system within AMUSE. Our main motivation was to see if we could replicate the Apep system in AMUSE, and from there discover more about its evolution and its fascinating morphology. This also includes attempting to make the nitrogen-burning star go supernova.

In section 2, we discuss how we modeled Apep in AMUSE, including its initial parameters, gravity and hydrodynamics. In section 3, we discuss our attempt at creating a supernova in Apep. In section 4, we show the results from our simulation, which will be discussed in section 5. The final conclusions will be stated in section 6.

2 Modeling Apep

To model Apep, we had to collect many observed/modelled parameters and set up the parameter space. After initializing the Apep system in AMUSE, we used gravity and hydrodynamical codes to evolve the model.

2.1 Initial parameters

The Apep system has three stars, one WR binary, and a companion supergiant. The binary has not been resolved, thus their masses can not be accurately determined. The enclosed mass is

$19 \pm 12 M_{\odot}$ (Han et al. 2020). Within this range, we opted for $30 M_{\odot}$ in our simulation, since the expected mass range is between 20 and $40 M_{\odot}$ (Han et al. 2020).

2.1.1 Primary Wolf-Rayet star

The primary WR star is a carbon-burning star and is the likely candidate for near-critical rotation (J. R. Callingham, Crowther, et al. 2020). The most important parameters of this star are given below (Palacio et al. 2022):

- **Type:** WC8
- **Mass:** $19 M_{\odot}$
- **Radius:** $6.3 R_{\odot}$
- **Effective temperature:** 60 kK
- **Mass-loss rate:** $2.9 \cdot 10^{-5} M_{\odot} yr^{-1}$
- **Terminal wind velocity:** 2100 km s^{-1}

2.1.2 Secondary Wolf-Rayet star

The secondary WR star is a nitrogen-burning star, which is the likely candidate for the super-nova. Its important parameters are given below (Palacio et al. 2022):

- **Type:** WN46b
- **Mass:** $11 M_{\odot}$
- **Radius:** $6 R_{\odot}$
- **Effective temperature:** 65 kK
- **Mass-loss rate:** $4 \cdot 10^{-5} M_{\odot} yr^{-1}$
- **Terminal wind velocity:** 3500 km s^{-1}

2.1.3 Supergiant star

Not a lot is known about the Northern companion, except its distance and orbital period around the Central Engine. We can use HD 151804 and He 3–759 (Crowther and Evans 2009) as a reference for an O8 Iaf supergiant. This gives us the following parameters:

- **Type:** O8-Iaf
- **Mass:** $40 M_{\odot}$
- **Radius:** $35.4 R_{\odot}$
- **Effective temperature:** 30 kK
- **Mass-loss rate:** $10^{-5} M_{\odot} yr^{-1}$

2.1.4 Binary orbital elements

From a geometric model of Apep, (Han et al. 2020) found the best-fitting parameters of the orbit to be:

- **Semi-major axis:** 67 au
- **Period:** 125 years

- **Eccentricity:** 0.7
- **Inclination:** $\pm 25^\circ$
- **Argument of periastron:** 0°

Note that in our code, we set the inclination to 0 degrees, since we wish to have the WR binary on the x-y plane for the rotation.

2.1.5 Companion orbital elements

The distance from the Central Engine to the companion was determined by using the distance to Apep from Earth, ~ 2.4 kpc (J. R. Callingham, Tuthill, et al. 2018).

- **Distance to companion:** 1700 au
- **Period:** 10^4 yr
- **Eccentricity:** 0 (guess)
- **Inclination:** 25° (guess)?
- **Longitude of ascending node:** 45° (guess)

2.2 Stellar orbits

To initialise Apep, we generate a binary from the orbital elements of the Central Engine (subsection 2.1), and set all initial parameters to the binary particles. After this, we generate another binary with the Central Engine and the supergiant, and add the supergiant particle to the inner binary particle set. The supergiant particle is also given all known parameters from subsection 2.1. The initial system is shown in Figure 2:

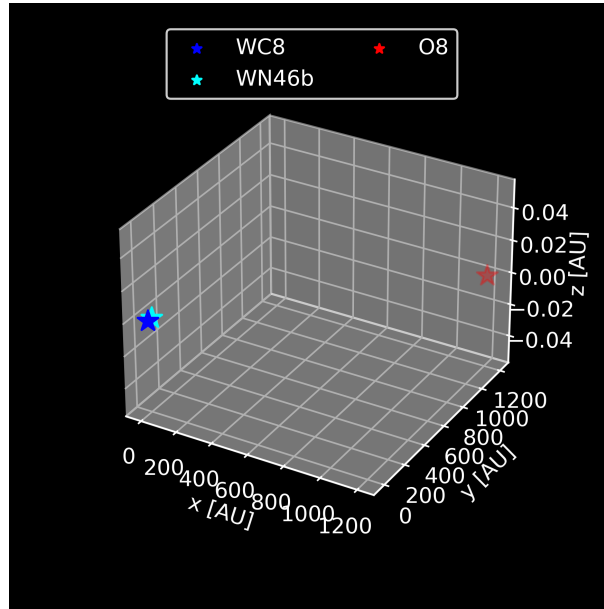


Figure 2: 3D projection of the initial Apep system.

2.3 Gravity

For the gravity calculations in our simulation, we used the **ph4** code, which is based on the fourth-order Hermite predictor–corrector scheme. This code allowed us to precisely follow the motions of the particles in the Apep system by adjusting only the time-step, without any extra physical parameters to tweak. We set up the N -body converter to use the total mass of the stars in Apep and the distance between the Central Engine and the supergiant. Other than that, we kept the gravity code as simple as possible.

2.4 Hydrodynamics

In our simulation, the hydrodynamic interactions of the stellar wind from the central engine were modeled using **Fi**, a TreeSPH type code. Fi treats each particle as if it is spread over a defined region, allowing for a fluid-like behavior that captures the complex gas dynamics within the system. We used Fi to model the supernova event as well.

2.5 Stellar wind

Initially, we tried to model the stellar wind using the `stellar_wind` routine in the AMUSE package, but very quickly, we realised that we could not make it work for our binary central engine, especially with the simple calculations of the gas particle’s internal energy that were needed by us in our initial plotting routine that simply plotted each SPH particle (without gravity code).

Therefore we adapted a routine to generate new SPH particles modeling the stellar wind based on the mass loss rate of the two Wolf-Rayet stars in the central engine of the Apep System.

2.5.1 Stellar rotation

At first, we simulated without any stellar rotation. However, as discussed in [section 1](#) the near-critical rotation of WC8 could be one of the main drivers for the pinwheel structure of Apep. Therefore, we added a tangential velocity of 500 km s^{-1} to the stellar wind particles launched from WC8’s surface.

2.6 Bridge

We created a bridge by constructing a gravhydro solver that combines the N -body code gravity and the hydrodynamics code hydro for the stellar winds. The diagram in [Figure 3](#) shows how the Bridge connects the N -body gravity code and SPH hydrodynamics code, and how the adapted stellar winds uses gravity to create SPH particles used for the hydrodynamics code:

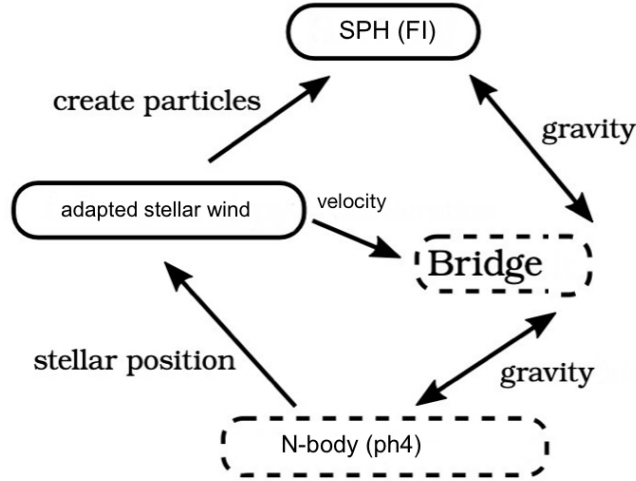


Figure 3: Flow chart of our Bridge and adapted stellar wind. Adapted from (Van Der Helm et al. 2019).

Once the bridge is created and the solver is added, we set the bridge timestep as the minimum between the gravity and hydro timesteps. The decision of timestep took many iterations because a longer timestep resulted in inaccurate coupling of the two solvers and a shorter timestep made the computation slow. There was a bug in **Fi** that broke the code for larger timesteps, which prevented us from getting any decent results for a long time. Steven Rieder proposed a fix (which worked (partly) for us) here: <https://github.com/amusecode/amuse/issues/1014>

2.7 Model evolution

We focused on the effects of the stellar winds from the two Wolf-Rayet stars in the central engine. To do this, we simulated for 150 years, matching the orbital period of these stars, to see how their movements and the stellar winds shape the system over time. In the model evolution, we added the particles in the central engine to the adapted stellar wind code and the mass of the SPH particles to create stellar wind particles. We made several channels between the particles and the gravity/hydro particles and copied these. Each selected timestep was saved to a .hdf5 file.

3 What if WN46b goes supernova?

In our simulation of the Apep system, we tried to explore the effects of a supernova event by focusing on the WN46b Wolf-Rayet star. We opted to simulate the WN46b star going supernova first due to its advanced evolutionary stage and the higher likelihood of it exploding before its companion in the Apep system, as suggested by its rapid nitrogen-burning process. Our initial approach involved two distinct scenarios: a more realistic one where we would let the Apep system naturally evolve to a longer time where we have the full morphology and an alternative one where we triggered the supernova prematurely in the system's early evolution.

3.1 Preliminary Supernova Simulation

Initially, we chose to simulate the supernova at an early stage. This was done to ideally allow us to test our setup and ensure that the supernova event was correctly implemented. Using MESA, we evolved the WN46b star for a million years, after which we converted its model into a pickle file and subsequently used it to make SPH particles to simulate the supernova. To do this, we injected high energies ($\sim 10^{51}$ erg) into the SPH particles.

4 Results and workflow

Our simulation ran the Apep system. In an attempt to replicate the morphology of the system, we have tried several different approaches and physical phenomena to achieve this. The first one is the Apep system with only their stellar winds; in the second approach we added a rotation to the WC8 star; and the final approach was designed to predict the future morphology of the Apep system with a supernova of the WN46b star. For every simulation we ran we produced figures for each timestep using our plotting routine and turned this into an animation using ffmpeg.

4.1 Preliminary results

Before our first approach, we tested the winds without any moving stars. For this, we used an example from (Portegies Zwart and McMillan 2018). This example used a function that transformed the mass lost due to stellar wind into sph particles that are sent isotropically outwards. See the result in Figure 4:

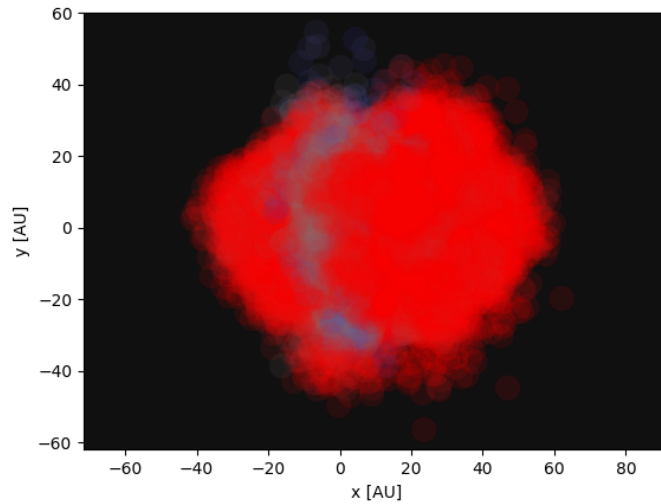


Figure 4: Snapshot at the final timestep of stellar wind only simulation. Also see the movie on https://github.com/sambeckers/Project-AMUSE-/blob/main/movies/hydro_central_engine.mp4

4.2 Without rotation

In our first approach, we tested the stellar winds bridging both gravity and hydro codes. Evolved for 7.25 years with a timestep of 0.01 years. See the result in Figure 5:

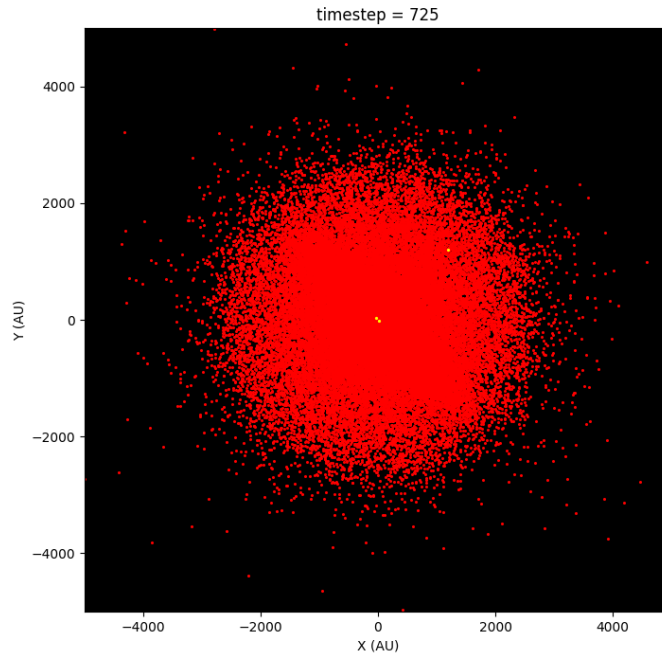


Figure 5: Snapshot of stellar wind simulation without rotation of star. Also see the movie on https://github.com/sambeckers/Project-AMUSE-/blob/main/movies/apec_.mp4

This image does not resemble the morphology of the Apec system. This could be the case because the simulation does not include every intricacy of the real Apec system. Star rotation was one of the suggested physical phenomena in (J. R. Callingham, Tuthill, et al. 2018).

4.3 With rotation

In our second approach, we implemented a rotation to the WC8 star at a near-critical velocity of 500 km s^{-1} . The rotation is assumed to be in the xy-plane. We added this velocity tangentially to the sph particles. Evolved for 9 years with a timestep of 0.01 years. See the result in Figure 6:

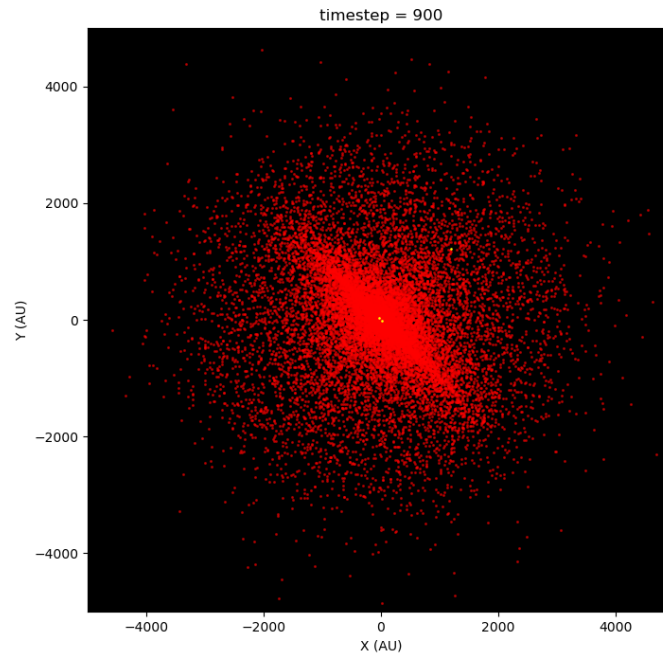


Figure 6: Snapshot of stellar wind simulation with stellar rotation.

This looks like the morphology of the Apep system. At a later stage, we uncovered a mistake in the code that did not give the right direction to the SPH particles. Adjusting for this mistake we evolved for 6.29 years with a timestep of 0.01 years. See the result in [Figure 7](#):

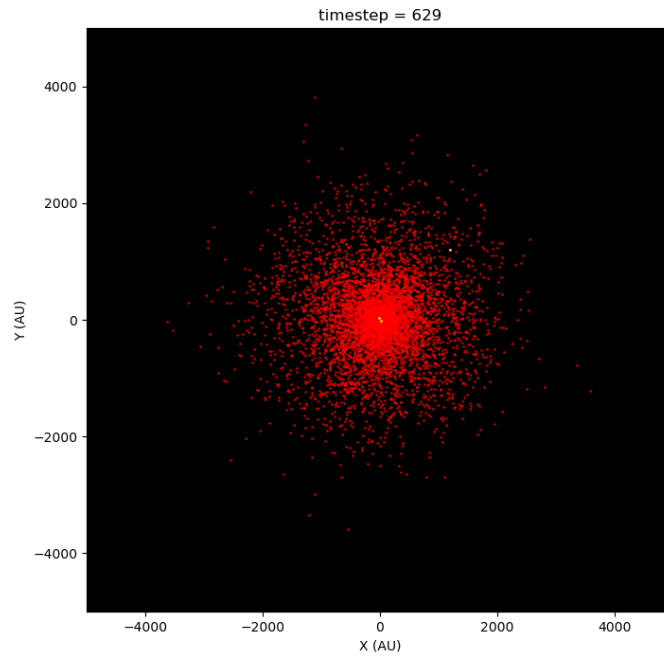


Figure 7: Snapshot of stellar wind simulation with stellar rotation. And bug fix.

This adjustment did not result in the aimed morphology. Puzzled why this happened, we tested the wind code with a single burst of sph particles at the start of the simulation. Here we converted the mass lost due to wind into a total of 10,000 sph particles divided among both stars. After 0.26 years and a timestep of 0.01 years. See the result in [Figure 8](#):

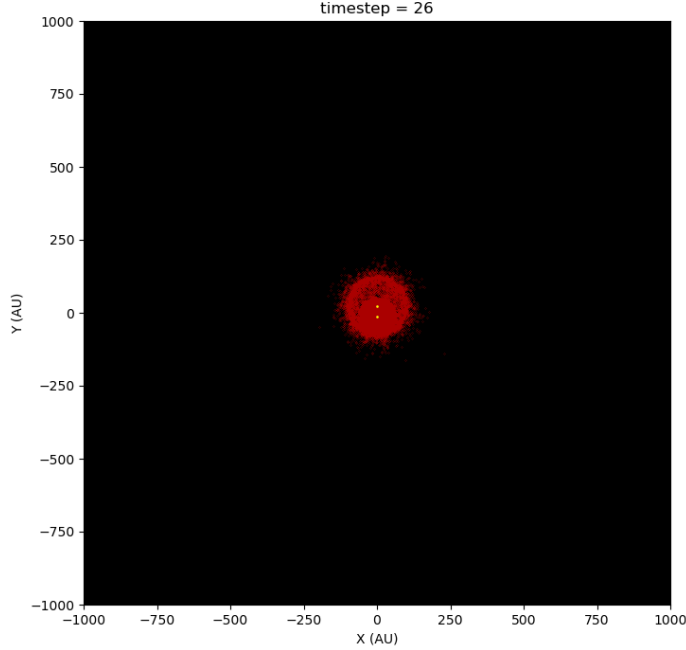


Figure 8: Snapshot of stellar wind simulation with a single burst of sph particles.

The resulting image does not give the impression that the wind particles collide. Since there does not seem to be a frontier of sph particles with equal distance to both stars.

5 Discussion

5.1 Without rotation

In our simulation, we observed no peculiar morphology characteristic of the Apep system; the results resembled a point source emitting SPH particles isotropically. It appeared that the hydrodynamics were not functioning as expected, and even with high resolution, we did not observe the anticipated collisions among particles.

5.2 With rotation

In one scenario, the rotational velocity was added in one direction (v_y) and subtracted in the other (v_x). This setup gave a rotational motion around an axis perpendicular to the plane of motion, as one component of the velocity is increased while the other is decreased by the rotational velocity. However, the rotation did not yield the anticipated morphology.

The addition of rotational velocity introduces an additional velocity component in both directions, which leads to a combined effect of rotation and flipping the direction due to how the rotational velocity is applied in the same direction to both velocity components. This was an erroneous case and the only case that gave a resembling morphology to the Apep System.

5.3 Challenges encountered with the supernova

The supernova was initiated by injecting SPH particles with high energies intending that these particles will also blow away the stellar wind particles and change the morphology of our system. On the contrary, these particles did not occupy the expected regions, even after adjusting the explosion energy. All the SPH particles created at the time of the supernova remained spherically intact around the core of the star [See figure [Figure 9](#)]. A kinetic energy test of the system before and after the supernova event did suggest that the SPH particles used to make the supernova have indeed gotten the injected energy. A possible reason for this discrepancy could be the timestep used for our simulation, ranging in a fraction of a year, was too large to capture the rapid developments of a supernova accurately. Therefore, we dropped our second scenario of letting the Apep system evolve for a longer time and see how the supernova would change its morphology.

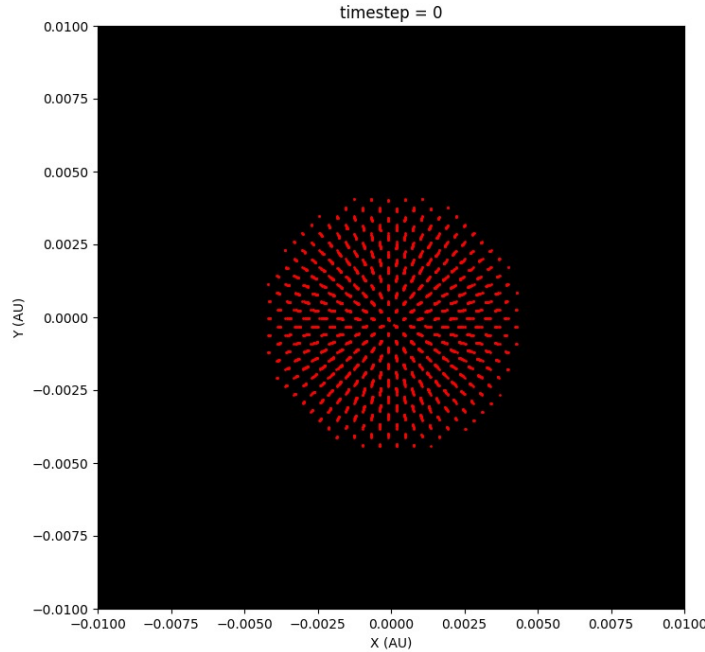


Figure 9: SPH particles injected with supernova energy

6 Conclusion

In conclusion, our simulation of the Apep system provided insightful yet challenging results. While we successfully modeled the system's basic structure and dynamics, replicating the unique morphology of Apep proved difficult. The addition of rotation and the attempt to simulate a supernova event highlighted the complexities of such phenomena. Our findings emphasise the importance of the bridge, its accurate time-stepping to capture details of SPH particles and rapid events like supernovae, and the need for further refinement in our simulation techniques. This shall contribute to any future efforts to understand the intricate behaviours of colliding wind binaries and their end-of-life events.

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